SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Deepak V-192411021 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

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Debugging and Optimizing IPv4 Network Configuration

Given,

IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192 (/26)

Default Gateway: 192.168.10.1

Solution,

Subnet Details:

-> Network Address: 192.168.10.64

-> Host Range: 192.168.10.65 – ~~to~~ 192.168.10.126

-> Broadcast Address: 192.168.10.127

Configuration Verification:

-> IPv address ~~is~~ 192.168.10.65 is Valid

-> ~~IRC~~ Default Gateway 192.168.10.1 is Invalid because it belongs to different subnet.

Optimized Configuration:

-> IP address: 192.168.10.65

-> Default Gateway: 192.168.10.66.

Any valid host address within same subnet configured as router interface.

2) Debugging IPv6 Address Configuration:

Given,

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Solution,

Expanded Addresses:

- Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix:

- Device A: 2001:db8:0:0::/64
- Device B: 2001:db8:0:0:1::/64

Configuration Verification:

→ The devices do not belong to the same subnet.

Optimized Configuration:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8::1/64

Optimizing Subnet Allocation Using VLSM:

Given,

- Network Address: 192.168.1.0/24
- Existing Subnets: /26, /27, /28

Solution,

Existing Subnets,

=> /26 : Network: 192.168.1.0;
Host Range: 192.168.1.1 -> 192.168.1.62
Broadcast: 192.168.1.63

=> /27 : Network: 192.168.1.64.
Host Range: 192.168.1.65 - 192.168.1.94
Broadcast: 192.168.1.95

=> /28 : Network: 192.168.1.96
Host Range: 192.168.1.97 - 192.168.1.110
Broadcast: 192.168.1.111

Debugging:

-> Some departments have unused IP addresses
-> Use VLSM to assign larger subnets.

4) Debugging IPv6 Address Configuration

Given,

- Device A: 2001:db8:1234::10/64
- Device B: 2001:db8:1235::20/64

Solution,

Expanded Addresses:

- Device A: 2001:0db8:1234:0000:0000:0000:0000:0010
- Device B: 2001:0db8:1235:0000:0000:0000:0000:0020

Network Prefix:

- Device A: 2001:db8:1234::/64
- Device B: 2001:db8:1235::/64

Configuration Verification:

- The devices belong to ~~g~~ different subnets.

Optimized Configuration:

- Device A: 2001:db8:1234::10/64
- Device B: 2001:db8:1234::20/64

Available Host Addresses:

- 2^{64} = 18,446,744,073,709,551,616

Debugging Router Forwarding using Routing Table.

Given,

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP: 192.168.2.100

Solution,

Longest prefix Match:

-> Destination IP 192.168.2.100 matches the route 192.168.2.0/24.

Selected Next-Hop Router:

-> The packet is forwarded to 10.0.0.3.

~~Desti~~ Destination Verification:

-> Network Address: 192.168.2.0/24.

-> Valid Host Range: 192.168.2.1 – 192.168.2.254

-> Broadcast Address: 192.168.2.255

-> 192.168.2.100 is a valid host.

Configuration Error:

→ If packets fail to reach the destination, the next hop router 10.0.0.3 may be unreachable.

Optimized Routing:

→ Verify connectivity of 10.0.0.3.

→ Ensure the Routing entry for 192.168.2.0/24,